

1.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A particle P moves on the x -axis.

At time t seconds, $0 \leq t \leq 3$, the displacement of P relative to the origin is x metres where

$$x = 2t^4 - 14t^3 + \frac{45}{2}t^2 + 14t$$

(a) Verify that P is at instantaneous rest when $t = 2$

(3)

Given that this is the only value of t , in the interval $0 \leq t \leq 3$, for which P is at rest,

(b) find the total distance travelled by P in the interval $0 \leq t \leq 3$

(2)

(c) Find the magnitude of the acceleration of P when $t = 1.5$

(3)

2. [In this question you may use, without proof, the formula for the centre of mass of a uniform sector of a circle.]

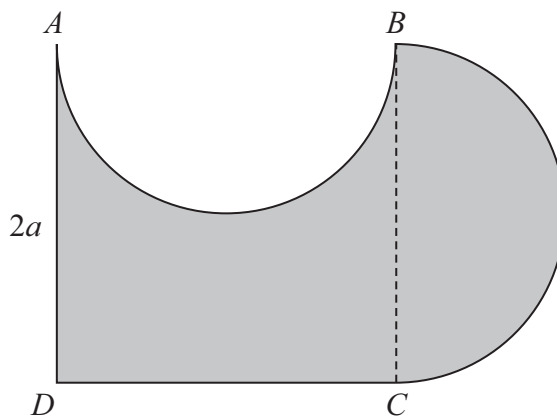


Figure 1

A uniform square lamina $ABCD$ has side $2a$.

A semicircular lamina, with diameter AB , is removed from the square lamina.

The semicircular lamina is then attached to BC to form the uniform lamina L shown shaded in Figure 1.

The centre of mass of L is a distance $\bar{x}$ from AD .

(a) Show that

$$\bar{x} = \frac{a}{24}(28 + 3\pi)$$

(5)

The lamina L has weight W .

A particle of weight kW is attached to L at D to form a loaded lamina.

The loaded lamina is now suspended by two light vertical strings with the first string attached at A and the second string attached at B .

The loaded lamina hangs freely in equilibrium with DC horizontal.

Given that the tension in the string attached at A is $4T$ and the tension in the string attached at B is T

(b) find the exact value of k .

(5)

3. [*In this question, the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane, $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upwards.*]

The point A is on horizontal ground.

A ball is projected from A with velocity $(8\mathbf{i} + 14\mathbf{j})\text{ m s}^{-1}$

The motion of the ball is modelled as that of a particle moving freely under gravity.

- (a) Find the maximum possible height of the ball above the ground.

(2)

- (b) Find the speed of the ball 2.4 s after leaving A .

(3)

The point B is also on the horizontal ground.

The point C is 3 m vertically above B .

After the ball reaches its maximum height, the ball passes through the point C .

- (c) Find the distance AB .

(5)

4. The combined mass of a cyclist and her cycle is 70 kg.

The cyclist travels on a straight horizontal road with constant acceleration 0.4 m s^{-2}

The resistance to the motion is modelled as a constant force of magnitude 30 N.

- (a) Using this model, find the rate at which the cyclist is working, at the instant when her speed is 5 m s^{-1}

(4)

The cyclist travels in a straight line **down** a hill from a point A to a point B , where $AB = 250 \text{ m}$

- at the instant she passes A , her speed is 5 m s^{-1} and she **stops** pedalling
- at the instant she passes B , her speed is 8 m s^{-1}
- the resistance to the motion is modelled as a constant force of magnitude 30 N

- (b) Use this model with the work-energy principle to find the change in vertical height from A to B .

(5)

The cyclist travels **up** a hill from a point C to a point D , where $CD = 200 \text{ m}$.

- she travels from C to D with constant speed, 8 m s^{-1}
- she travels in a straight line at 5° to the horizontal
- the resistance to the motion from non-gravitational forces is modelled as a constant force of magnitude 30 N

- (c) Use this model to find the total work done by the cyclist as she moves from C to D .

(3)

5. A particle P of mass 0.5 kg is moving with velocity $6\mathbf{i}\text{ m s}^{-1}$ in the xy plane.

The particle receives an impulse $(a\mathbf{i} + b\mathbf{j})\text{ N s}$, where a and b are constants.

As a result of the impulse, the direction of motion of P is deflected through 45°

The impulse has magnitude $\frac{3\sqrt{2}}{2}\text{ N s}$.

Find the two possible impulses, giving your answers in terms of $\mathbf{i}$ and $\mathbf{j}$.

(8)

6.

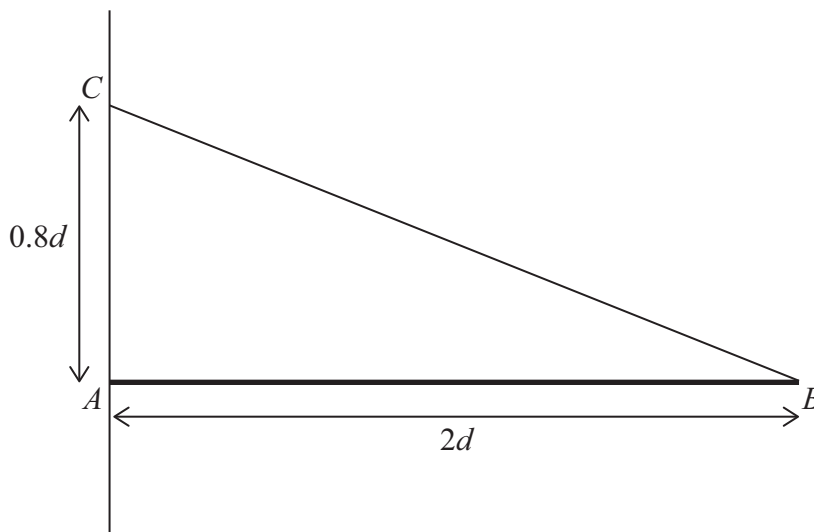


Figure 2

A uniform rod AB has mass m and length $2d$.

The rod is held in equilibrium in a horizontal position, with the end A against a rough vertical wall, by a light inextensible string.

One end of the string is attached to the rod at B .

The other end of the string is attached to the wall at the point C , where $AC = 0.8d$ and C is vertically above A , as shown in Figure 2.

A particle of mass km is attached at B and the rod AB remains in equilibrium in a horizontal position.

The particle, rod and string are in a vertical plane perpendicular to the wall.

(a) Show that the tension in the string is

$$\frac{mg\sqrt{29}}{4}(2k + 1) \quad (4)$$

The resultant force acting on the rod at A makes an angle θ with the horizontal.

(b) Given that $\tan \theta = \frac{1}{8}$, find the exact value of k . (6)

7.

**Figure 3**

Three particles, P , Q and R , lie at rest on a smooth horizontal surface.

The particles are in a straight line with Q between P and R , as shown in Figure 3.

The mass of P is $2m$, the mass of Q is $3m$ and the mass of R is $5m$.

The coefficient of restitution between each pair of particles is e .

Particle P is projected with speed $2u$ towards Q and they collide.

(a) Show that the speed of Q immediately after the collision is

$$\frac{4u}{5}(e + 1) \quad (5)$$

In the collision, the direction of motion of P is reversed.

(b) Find the full range of possible values of e . (3)

In the collision, the magnitude of the impulse exerted on Q by P is $\frac{108}{25} mu$

(c) Show that $e = \frac{4}{5}$ (3)

(d) Determine, with clear reasoning, whether there will be a second collision between P and Q . (6)